

Homework 10

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Collaboration Statement

I worked on this homework independently. No outside collaborators or resources were used.

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Problem 1 (*Ross Chapter 3 18.5 (a)*)

Let f and g be continuous functions in $[a, b]$ such that $f(a) \geq g(a)$ and $f(b) \leq g(b)$. Prove $f(x_0) = g(x_0)$ for at least one x_0 in $[a, b]$.

Discussion.

We want a point where $f(x_0) = g(x_0)$. This is equivalent to $f(x_0) - g(x_0) = 0$, so it is natural to consider the difference $h(x) = f(x) - g(x)$. Since f and g are continuous, h is continuous on $[a, b]$. The inequalities $f(a) \geq g(a)$ and $f(b) \leq g(b)$ then say $h(a) \geq 0$ and $h(b) \leq 0$, so 0 lies between $h(a)$ and $h(b)$. By the Intermediate Value Theorem applied to h , there is some $x_0 \in [a, b]$ with $h(x_0) = 0$, i.e. $f(x_0) = g(x_0)$.

Proof.

Define $h : [a, b] \rightarrow \mathbb{R}$ by

$$h(x) = f(x) - g(x).$$

Since f and g are continuous on $[a, b]$ and the difference of two continuous functions is continuous, it follows that h is continuous on $[a, b]$.

From the hypotheses we have

$$f(a) \geq g(a) \implies h(a) = f(a) - g(a) \geq 0,$$

and

$$f(b) \leq g(b) \implies h(b) = f(b) - g(b) \leq 0.$$

Thus 0 lies between $h(a)$ and $h(b)$; equivalently, $h(b) \leq 0 \leq h(a)$.

By the Intermediate Value Theorem (Theorem 18.2 in Ross), since h is continuous on $[a, b]$ and 0 lies between $h(a)$ and $h(b)$, there exists some $x_0 \in [a, b]$ such that

$$h(x_0) = 0.$$

By the definition of h this means

$$f(x_0) - g(x_0) = 0,$$

so $f(x_0) = g(x_0)$.

Therefore there exists at least one $x_0 \in [a, b]$ with $f(x_0) = g(x_0)$, as required. $\square$

Problem 2 (*Ross Chapter 3 18.10*)

Suppose f is continuous on $[0, 2]$ and $f(0) = f(2)$. Prove there exist x, y in $[0, 2]$ such that $|y - x| = 1$ and $f(x) = f(y)$. *Hint:* Consider $g(x) = f(x + 1) - f(x)$ on $[0, 1]$.

Discussion.

Proof.

□

Problem 3 (*Ross Chapter 3 19.6 (b)*)

(From Part (a):) Let $f(x) = \sqrt{x}$ for $x \geq 0$. Show f' is unbounded on $(0, 1]$ but f is nevertheless uniformly continuous on $(0, 1]$ Compare with Theorem 19.6.

Show that f is uniformly continuous on $[1, \infty)$.

Discussion.

Proof.

□

Problem 4 (*Ross Chapter 3 19.7*)

(a) Let f be a continuous function on $[0, \infty)$. Prove that if f is uniformly continuous on $[k, \infty)$ for some k , then f is uniformly continuous on $[0, \infty)$.

(b) Use (a) and Exercise 19.6(b) to show that $f(x) = \sqrt{x}$ is uniformly continuous on $[0, \infty)$.

Discussion.

Proof.

□

Problem 5 (*Uniform Continuity of f*)

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$. Prove that if there exists a real number $M \geq 0$ such that $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in \mathbb{R}$, then f is uniformly continuous on $\mathbb{R}$.

Discussion.

Proof.

□

Problem 6 (*Differentiability of f*)

Suppose that $0 \in (a, b)$ and $f(x) = xg(x)$ for some function $g : (a, b) \rightarrow \mathbb{R}$ which is continuous at $x = 0$. Prove that f is differentiable at $x = 0$, and find $f'(0)$ in terms of g .

Hint: Note that you cannot use the product rule, since $g(x)$ is not guaranteed to be differentiable at $x = 0$

Discussion.

Proof.

□